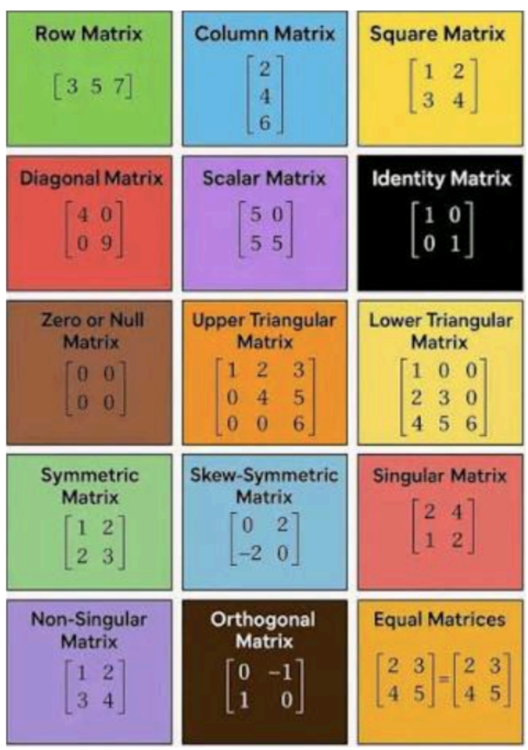
MATH 284 MATLAB Project #2 - Matrix Algebra Workbench



Project Objective

Develop a MATLAB application that allows users to create, manipulate, analyze, and visualize matrices while exploring the major concepts of matrix algebra.

Instead of only computing solutions, the program explains matrix properties and whether a matrix satisfies important linear algebra conditions.

Concepts And Textbook Chapters/Sections Covered

Chapter 2 - Matrix Algebra

- 2.1 Matrix Operations
- 2.2 Inverse of a Matrix
- 2.3 Characterizations of Invertible Matrices
- 2.8 Subspaces of R^n
- 2.9 Dimension and Rank

- Matrix operations
- Matrix inverse
- Invertible Matrix Theorem
- Subspaces of R^n
- Dimension
- Rank

GUI Layout

```
-------------------------------------------------------------
|              Matrix Algebra Workbench                    |
-------------------------------------------------------------
| Matrix A                      Matrix B                   |
| [              ]              [             ]            |
|                                                      |
| Rows: []   Columns: []                                 |
-------------------------------------------------------------
| Matrix Operations                                    |
|                                                      |
| [+] [-] [Multiply] [Transpose]                       |
| [Inverse] [Determinant] [Rank]                       |
-------------------------------------------------------------
| Matrix Analysis                                      |
|                                                      |
| Is Invertible?                                       |
| Rank                                                 |
| Nullity                                              |
| Dimension                                            |
| Independent Columns?                                 |
-------------------------------------------------------------
| Visualization                                        |
|                                                      |
| Column Space / Transformation / Basis Display        |
-------------------------------------------------------------
| Advanced Tools                                       |
|                                                      |
| [Invertible Matrix Test]                             |
| [Subspace Explorer]                                  |
| [Column Space]                                       |
| [Row Space]                                          |
| [Null Space]                                         |
-------------------------------------------------------------
```

Necessary MATLAB Files For This Project

MatrixAlgebraWorkbenchApp.m

Responsibilities:

- ***Build GUI***
- ***Handle user interaction***
- ***Display results***
- ***Connect all analysis modules***

MatrixAnalyzer.m

Stores the following information about the matrix:

- ***inverse***
- ***determinant***
- ***rank***
- ***dimension***

InvertibilityChecker.m

Checks whether

- ***determinant ≠ 0***
- ***inverse exists***
- ***full rank***
- ***pivot in every row***
- ***pivot in every column***
- ***columns are linearly independent***
- ***unique solution for Ax=b***

SubspaceExplorer.m

Computes:

- *Column Space*
- *Row Space*
- *Null Space*

RankDimensionAnalyzer.m
Computes:

- *Rank*
- *Nullity*
- *Dimension*

This is used for the Rank-Nullity Theorem.

MatrixVisualizer.m
Responsible for visualization.

Necessary MATLAB Methods For This Project

createComponents()
performOperation()
analyzeMatrix()
displayResults()
clearWorkspace()
add()
subtract()
multiply()
transpose()
inverse()
calculateRank()
calculateDeterminant()
isInvertible()
findColumnSpace()
findRowSpace()
findNullSpace()
plotBasis()
plotColumnVectors()
plotTransformation()
plotSubspace()

GUI Functions

- Add, subtract, multiply, transpose
- Compute inverse and verify $AA^{-1} = I$
- Interactive invertibility checker
- Column, row, and null space explorer
- Rank and dimension calculator

Pseudocode

START PROGRAM

Create GUI

Initialize:

  Matrix A

  Matrix B

  Results panel

  Plot panel

WHEN user selects Matrix Operation

  Read matrices

  Determine operation

  Perform calculation

  Display result

WHEN Analyze Matrix selected

```
    Compute determinant

    Compute inverse

    Compute rank

    Check invertibility

    Display all properties

WHEN Subspace Explorer selected

    Compute:

        Column Space

        Row Space

        Null Space

    Display basis

    Plot subspace (if possible)

WHEN Rank Tool selected

    Compute rank

    Compute nullity

    Display Rank-Nullity relationship

WHEN Clear selected

    Reset GUI
```

END PROGRAM